

Relevance Feedback

Introduction to Information Retrieval

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Motivation

- Searching depends on matching keywords between user-query and document.
- Searchers and document creators may use different keywords to denote same “concept”.
- Vocabulary mismatch $\implies$ poor retrieval quality.
- Users may not know what they are looking for, but theyll know when they see it.
- Boost recall: “find me similar documents...”

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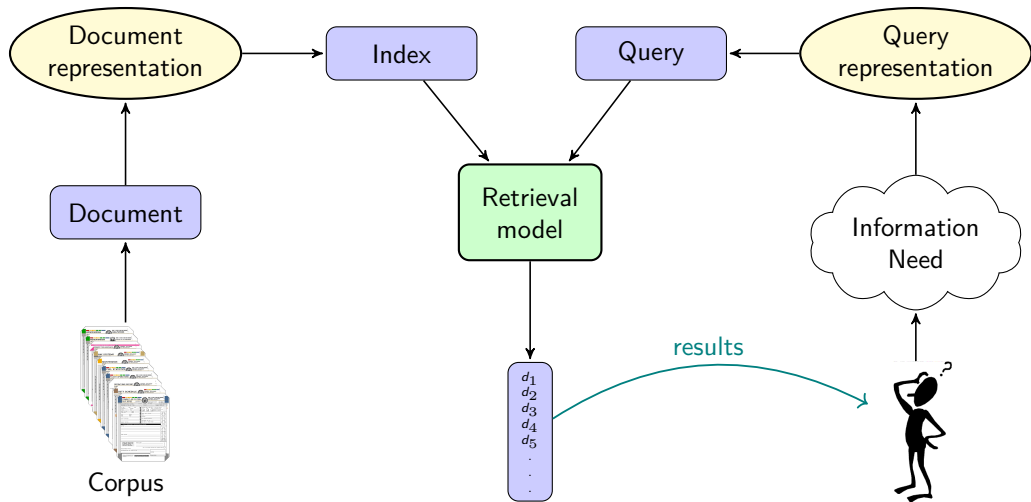
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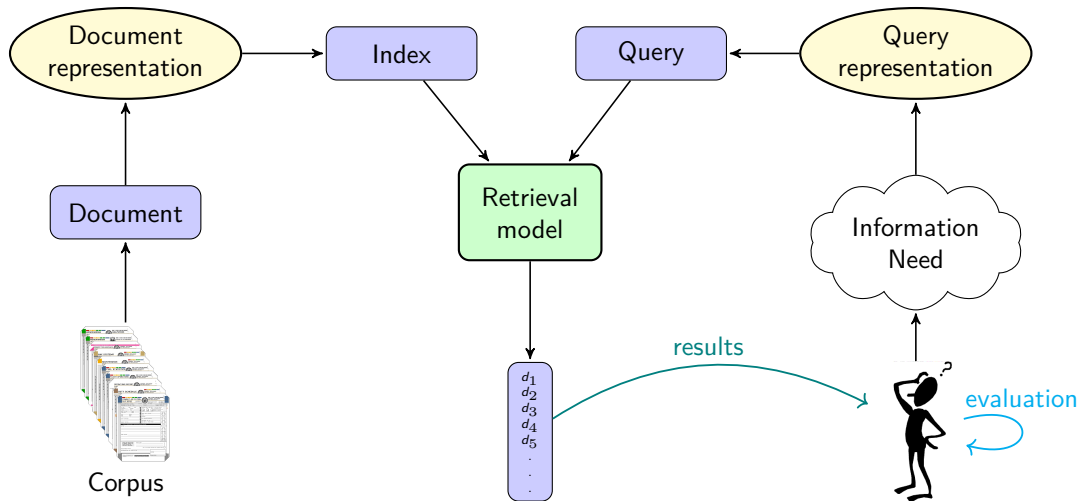
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- Solution: try to make a **better representation** of the information need.
 - ▶ **expand the query** by adding related words/phrases.
- Issues:
 - ▶ **select** which terms to add to query
 - ▶ **calculate weights** for added terms

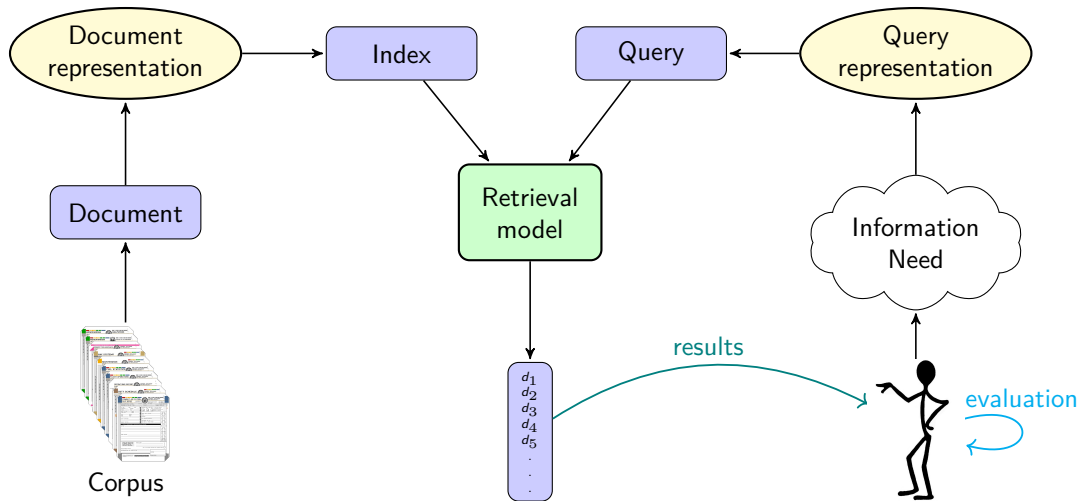
Relevance Feedback: A Graphical Representation



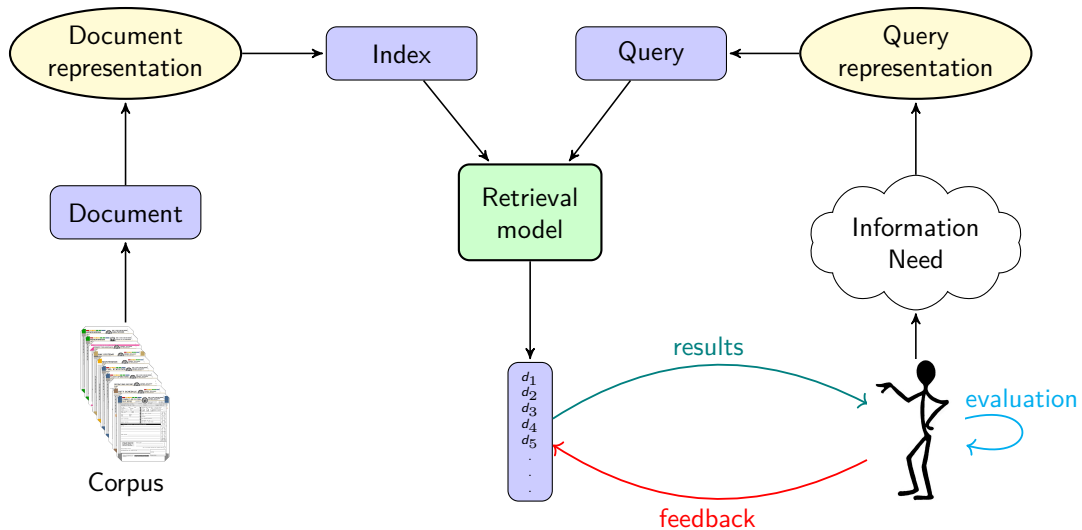
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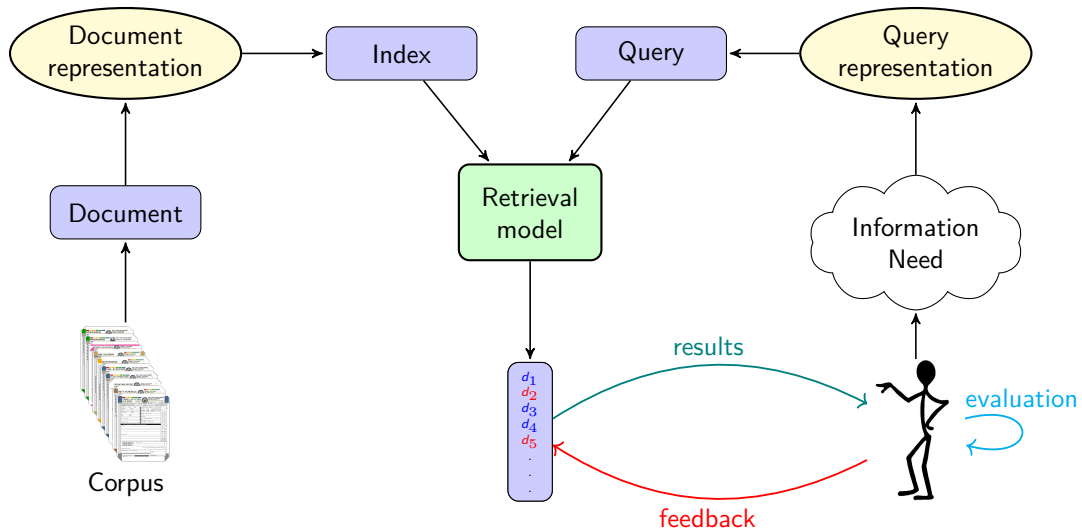
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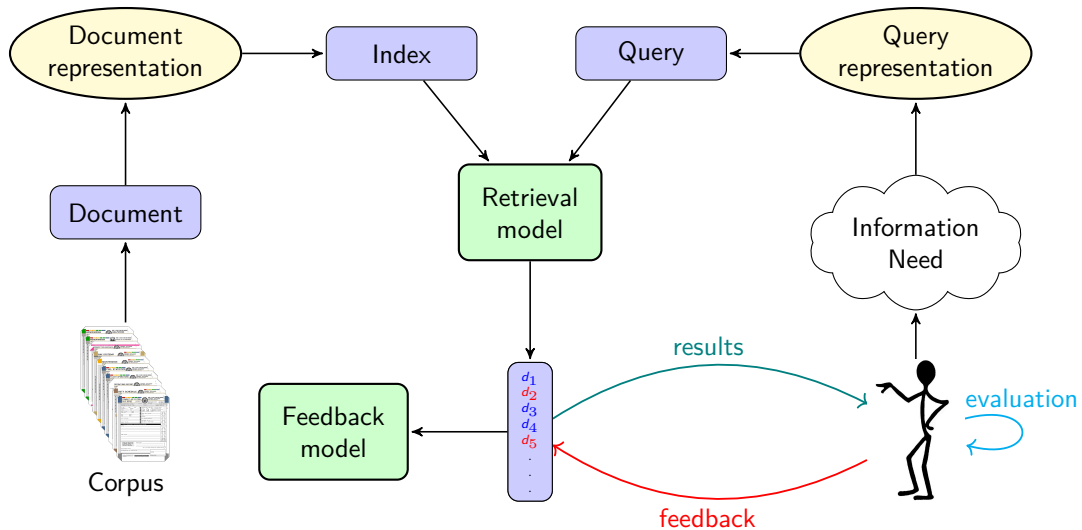
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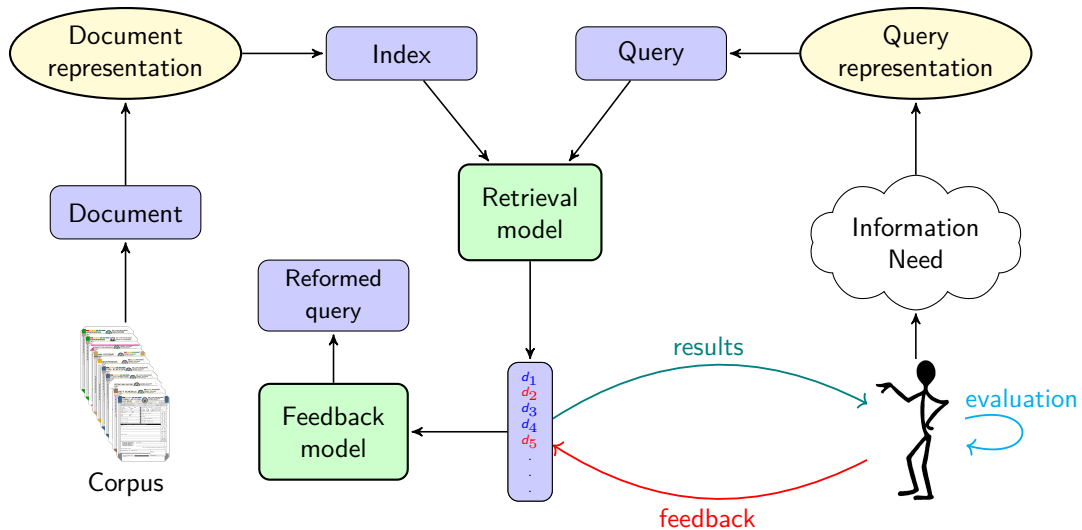
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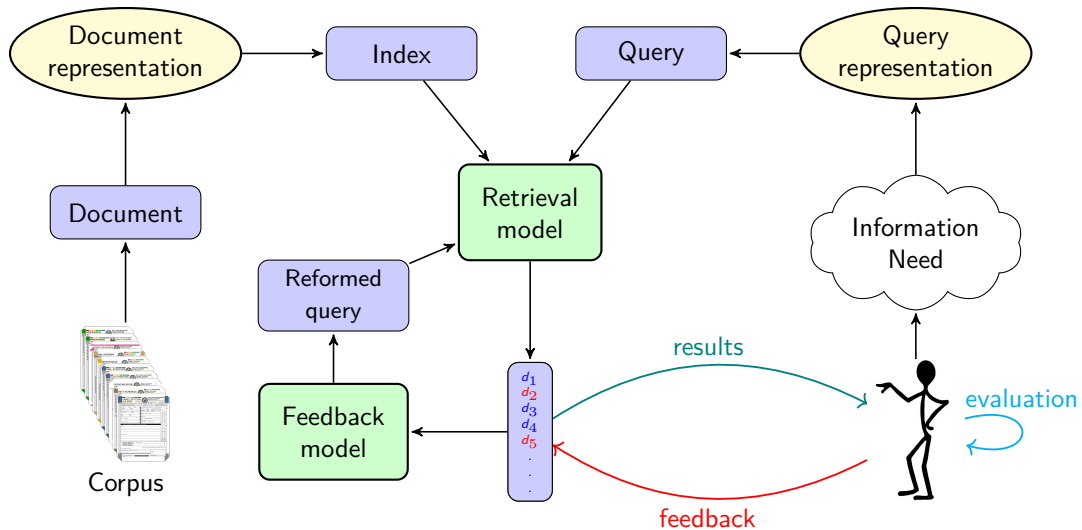
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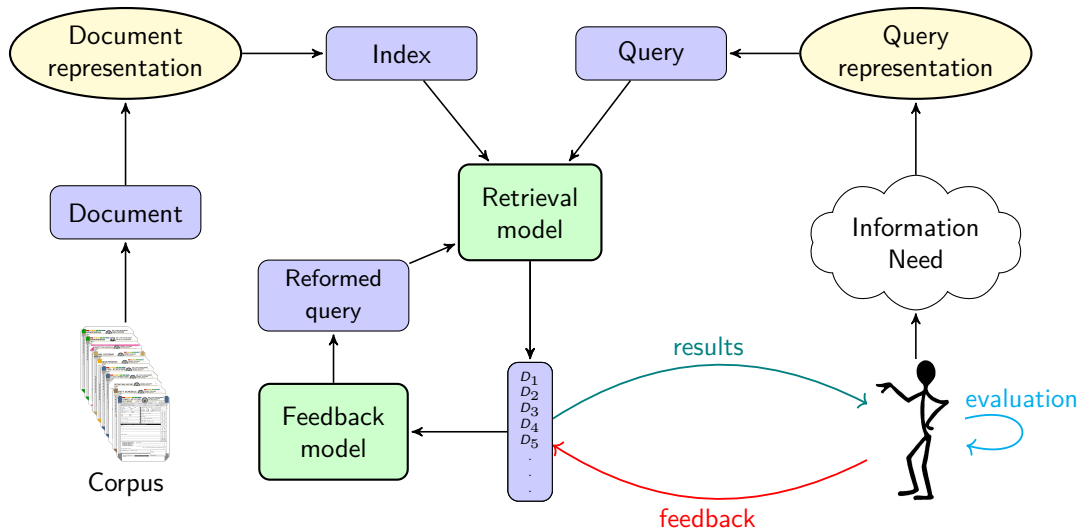
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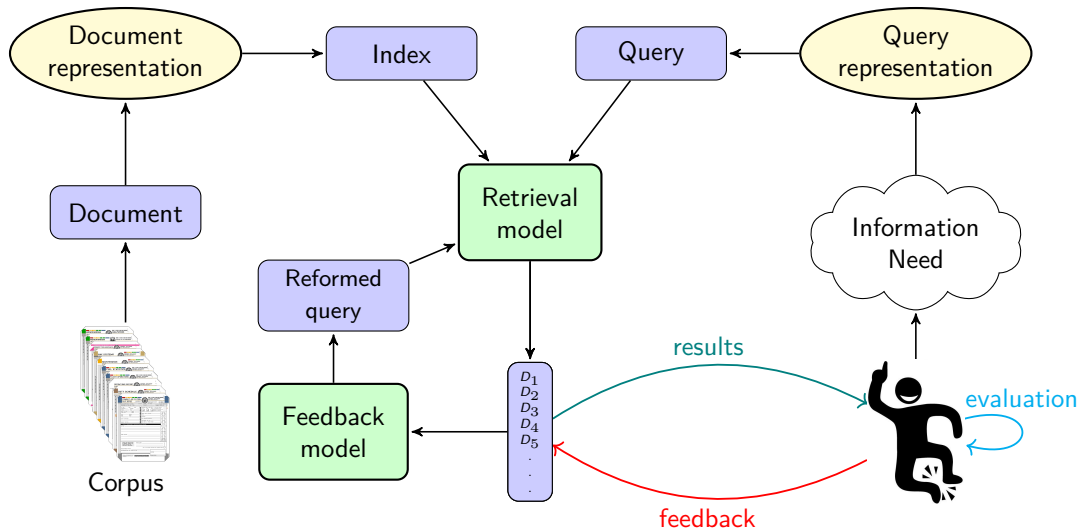
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Relevance Feedback: A Graphical Representation



Relevance Feedback: Basic Idea

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- Search engine performs retrieval with the reformulated query.
- The new set of documents returned by the engine, having (hopefully) better recall.

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- Can iterate this: several rounds of relevance feedback.

The Formal Relevance Feedback Problem

Given

- Original query vector $\mathbf{q}$
- Set of relevant documents D_r (marked by user)
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Key Assumptions

- ① User's initial query is a **reasonable starting point**
- ② Judged documents are **representative** of relevant/non-relevant sets

Explicit vs Implicit Feedback

Two kinds of user feedback

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- Explicit feedback is **more reliable but costly** to obtain.

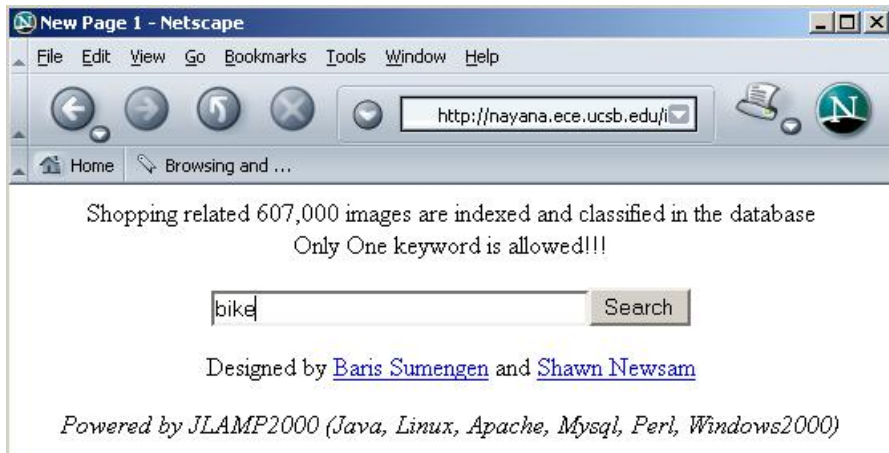
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- Explicit feedback is **more reliable but costly** to obtain.
- Real-world search engines **rely heavily on implicit** feedback.

Relevance Feedback: Example 1

- Initial Query: bike



Relevance Feedback: Example 1

- Results for initial query

Interface showing search results for an initial query, displaying a grid of images and their associated relevance scores.

Buttons: Browse, Search, Prev, Next, Random

					
(144473, 16458)	(144457, 252140)	(144456, 262857)	(144456, 262863)	(144457, 252134)	(144483, 265154)
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
					
(144483, 264644)	(144483, 265153)	(144518, 257752)	(144538, 525937)	(144456, 249611)	(144456, 250064)
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0

Relevance Feedback: Example 1

- Feedback: Relevant document selected

Interface showing a grid of 12 images related to bicycles and motorcycles, with a feedback mechanism. The interface includes buttons: Browse, Search, Prev, Next, Random.

					
(144473, 16458)	(144457, 252140)	(144456, 262857)	(144456, 262863)	(144457, 252134)	(144483, 265154)
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
					
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0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0

Relevance Feedback: Example 1

- Result after re-retrieval

[Browse](#) [Search](#) [Prev](#) [Next](#) [Random](#)

 (144538, 523493) 0.54182 0.231944 0.309876	 (144538, 523835) 0.56319296 0.267304 0.295889	 (144538, 523529) 0.584279 0.280881 0.303398	 (144456, 253569) 0.64501 0.351395 0.293615	 (144456, 253568) 0.650275 0.411745 0.23853	 (144538, 523799) 0.66709197 0.358033 0.309059
 (144473, 16249) 0.6721 0.393922 0.278178	 (144456, 249634) 0.675018 0.4639 0.211118	 (144456, 253693) 0.676901 0.47645 0.200451	 (144473, 16328) 0.700339 0.309002 0.391337	 (144483, 265264) 0.70170796 0.36176 0.339948	 (144478, 512410) 0.70297 0.469111 0.233859

Relevance Feedback: Example 2

- TREC 2 Topic - 113 : *New Space Satellite Applications*

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Rank	Score	Document title
1	0.539	NASA Hasnt Scrapped Imaging Spectrometer
2	0.533	NASA Scratches Environment Gear From Satellite Plan
3	0.528	Science Panel Backs NASA Satellite Plan, But Urges Launches of Smaller Probes
4	0.526	A NASA Satellite Project Accomplishes Incredible Feat: Staying Within Budget
5	0.525	Scientist Who Exposed Global Warming Proposes Satellites for Climate Research
6	0.524	Report Provides Support for the Critics of Using Big Satellites to Study Climate
7	0.516	Arianespace Receives Satellite Launch Pact From Telesat Canada
8	0.509	Telecommunications Tale of Two Companies

Relevance Feedback: Example 2

- TREC 2 Topic - 113 : *New Space Satellite Applications*

	Rank	Score	Document title
✓	1	0.539	NASA Hasnt Scrapped Imaging Spectrometer
✓	2	0.533	NASA Scratches Environment Gear From Satellite Plan
	3	0.528	Science Panel Backs NASA Satellite Plan, But Urges Launches of Smaller Probes
	4	0.526	A NASA Satellite Project Accomplishes Incredible Feat: Staying Within Budget
	5	0.525	Scientist Who Exposed Global Warming Proposes Satellites for Climate Research
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✓	8	0.509	Telecommunications Tale of Two Companies

Relevance Feedback: Example 2

- Initial query: *new space satellite applications*
- Expanded query after relevance feedback

2.074	new	15.106	space
30.816	satellite	5.660	application
5.991	nasa	5.196	eos
4.196	launch	3.972	aster
3.516	instrument	3.446	arianespace
3.004	bundespost	2.806	ss
2.790	rocket	2.053	scientist
2.003	broadcast	1.172	earth
0.836	oil	0.646	measure

Relevance Feedback: Example 2

- TREC 2 Topic - 113 : *New Space Satellite Applications*

Rank	Score	Document title
1	0.513	NASA Scratches Environment Gear From Satellite Plan
2	0.500	NASA Hasnt Scrapped Imaging Spectrometer
3	0.493	When the Pentagon Launches a Secret Satellite, Space Sleuths Do Some Spy Work of Their Own
4	0.493	NASA Uses Warm Superconductors For Fast Circuit
5	0.492	Telecommunications Tale of Two Companies
6	0.491	Soviets May Adapt Parts of SS-20 Missile For Commercial Use
7	0.490	Gaping Gap: Pentagon Lags in Race To Match the Soviets In Rocket Launchers
8	0.490	Rescue of Satellite By Space Agency To Cost \$90 Million

Relevance Feedback: Example 2

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Old	Rank	Score	Document title
2	1	0.513	NASA Scratches Environment Gear From Satellite Plan
1	2	0.500	NASA Hasnt Scrapped Imaging Spectrometer
	3	0.493	When the Pentagon Launches a Secret Satellite, Space Sleuths Do Some Spy Work of Their Own
8	4	0.493	NASA Uses Warm Superconductors For Fast Circuit
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Why Relevance Feedback Works

Statistical Intuition

- Terms that occur **frequently in relevant documents** but **rarely in non-relevant ones** have high discriminative value.
- Updating the query moves it closer to the cluster of relevant documents in vector space.
- Helps overcome the **vocabulary mismatch problem** by learning contextual or synonymous terms.

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- Updating the query moves it closer to the cluster of relevant documents in vector space.
- Helps overcome the **vocabulary mismatch problem** by learning contextual or synonymous terms.
- Especially useful when the initial query is short or ambiguous.

Query Expansion vs Query Reweighting

Two Reformulation Strategies

Query Expansion

- Add related terms.
- Example: “*bike*” → “*bicycle*”, “*cycling*”

Query Reweighting

- Adjust weights of existing query terms.
- Example: “*satellite ()*, *application ()*”

What is a Centroid?

- The centroid is the **center of mass** of a set of points
- In vector space model: documents = points in high-dimensional space

Mathematical Definition

For a set of documents D :

$$\mu(D) = \frac{1}{|D|} \sum_{d \in D} \mathbf{d}$$

Example

- Document 1: $\mathbf{d}_1 = [2, 4, 1]$
- Document 2: $\mathbf{d}_2 = [4, 2, 3]$
- Centroid: $\mu = \frac{[2,4,1] + [4,2,3]}{2} = [3, 3, 2]$

Rocchio's Algorithm for Relevance Feedback

- Standard algorithm used for relevance feedback (inside SMART, by *J.J. Rocchio*, in 1971).
- Integrates a measure of relevance feedback into Vector Space Model.
- **Maximize the difference** between average similarities for relevant and non-relevant documents.

Rocchio's Algorithm for Relevance Feedback

- Chooses the optimal query Q_{opt} that maximizes:

$$Q_{opt} = \operatorname{argmax}_Q \left[\textcolor{blue}{sim}(Q, \mu(D_R)) - \textcolor{red}{sim}(Q, \mu(D_{NR})) \right]$$

- D_R : Set of relevant documents.
- D_{NR} : Set of non-relevant documents.
- $\mu(D_X)$: Centroid of a set of documents X .

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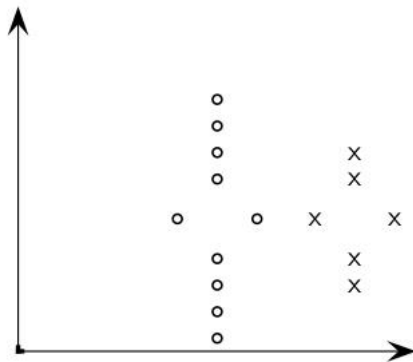
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To find the vector Q_{opt} , that separates **relevant** and **non-relevant** documents **maximally**.

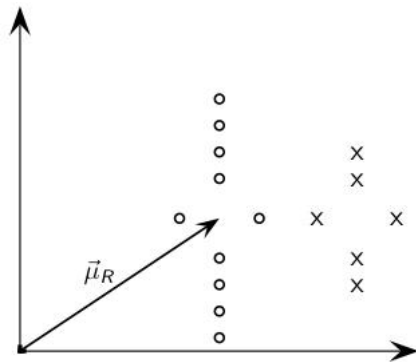
Rocchio's Algorithm Illustrated



○ relevant documents

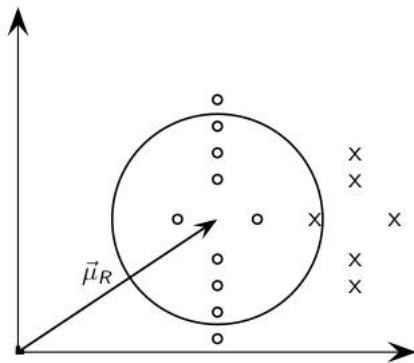
× non-relevant documents

Rocchio's Algorithm Illustrated



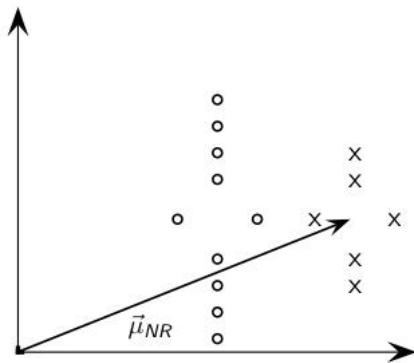
μ_R : centroid of relevant documents

Rocchio's Algorithm Illustrated



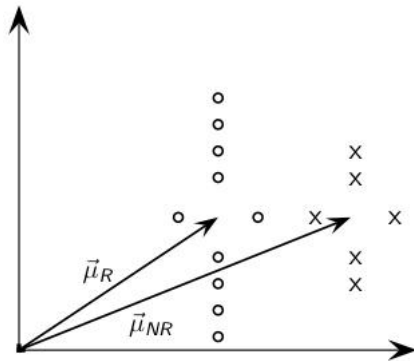
μ_R not separating relevant documents from non-relevant

Rocchio's Algorithm Illustrated

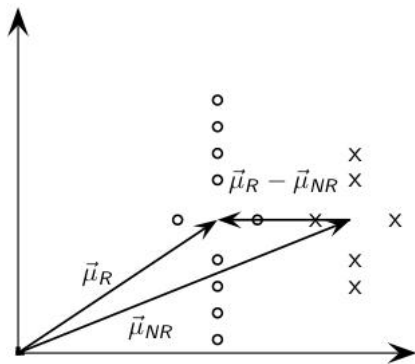


μ_{NR} : centroid of non-relevant documents

Rocchio's Algorithm Illustrated

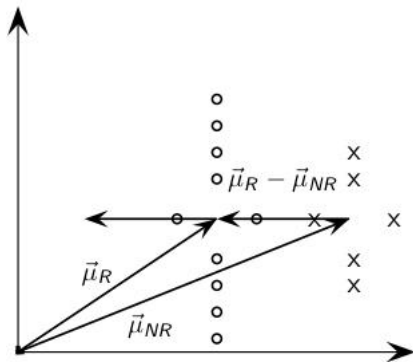


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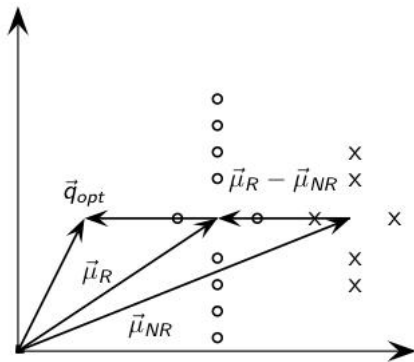


$\mu_R - \mu_{NR}$: difference vector

Rocchio's Algorithm Illustrated

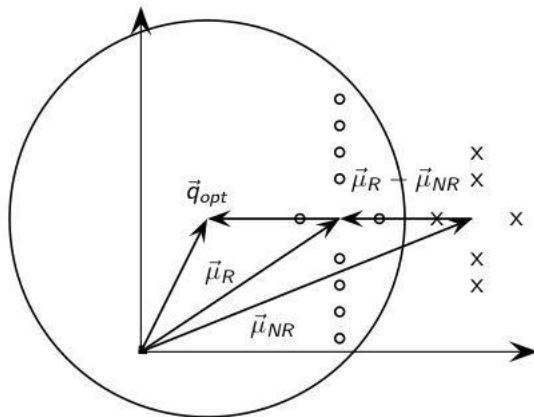


Rocchio's Algorithm Illustrated

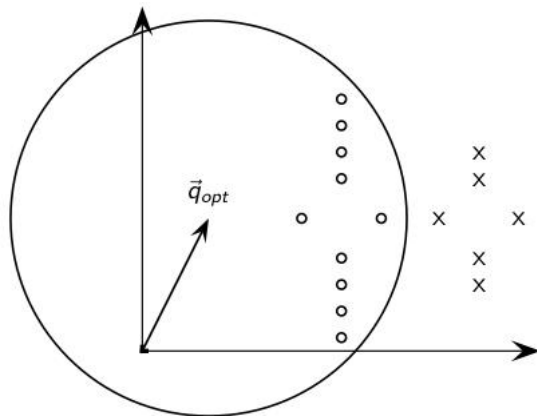


Add difference vector to μ_R to get μ_{opt}

Rocchio's Algorithm Illustrated



Rocchio's Algorithm Illustrated



$\vec{q}_{opt}$ separates relevant and non-relevant documents perfectly

Rocchio's Algorithm for Relevance Feedback

- The optimal query vector:

$$\begin{aligned} \mathbf{Q}_{opt} &= \mu(D_R) + \left[\mu(D_R) - \mu(D_{NR}) \right] \\ &= \frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i + \left[\frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i - \frac{1}{|D_{NR}|} \sum_{\mathbf{d}_j \in D_{NR}} \mathbf{d}_j \right] \end{aligned}$$

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The centroid of the **relevant documents** moved by the difference between the two centroids.

Rocchio's Algorithm

- In practice:

$$\mathbf{Q}_m = \alpha \mathbf{Q}_O + \beta \frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i - \gamma \frac{1}{|D_{NR}|} \sum_{\mathbf{d}_j \in D_{NR}} \mathbf{d}_j$$

- $\mathbf{Q}_O$: original query
- $\mathbf{Q}_m$: modified query
- D_R : set of known relevant documents
- D_{NR} : set of known non-relevant documents
- α, β, γ : weights (belief) of corresponding sets

Rocchio's Algorithm: Parameter Settings

$$\mathbf{Q}_m = \alpha \mathbf{Q}_O + \beta \frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i - \gamma \frac{1}{|D_{NR}|} \sum_{\mathbf{d}_j \in D_{NR}} \mathbf{d}_j$$

New (modified) query

- Set higher β, γ if there is a lot of judged documents.

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New (modified) query

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- Can set negative term weight to 0 (negative weight doesn't make sense in vector space model).

Rocchio's Algorithm: Example

$$\mathbf{Q}_m = \alpha \mathbf{Q}_O + \beta \frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i - \gamma \frac{1}{|D_{NR}|} \sum_{\mathbf{d}_j \in D_{NR}} \mathbf{d}_j$$

Original query

0	4	0	8	0	0
---	---	---	---	---	---

$\alpha = 1.0$

0	4	0	8	0	0
---	---	---	---	---	---

Positive feedback

2	4	8	0	0	2
---	---	---	---	---	---

$\beta = 0.5$

1	2	4	0	0	1
---	---	---	---	---	---

Negative feedback

8	0	4	4	0	16
---	---	---	---	---	----

$\gamma = 0.25$

2	0	1	1	0	4
---	---	---	---	---	---

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New query

-1	6	3	7	0	-3
----	---	---	---	---	----

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$\gamma = 0.25$

2	0	1	1	0	4
---	---	---	---	---	---

New query

-1	6	3	7	0	-3
----	---	---	---	---	----

We are performing **Query Expansion**.

Rocchio's Algorithm: issues

What should be regarded as *NR*?

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- Rank terms by their weights.
- Add 50-100 terms.

Explicit Relevance Feedback: Problem

- Usually Expensive with respect to time.
- Users are reluctant to provide explicit feedback.

Implicit Relevance Feedback

- Uses *Evidences* rather than *Explicit* feedback.

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Implicit Relevance Feedback

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- Not specific to users.
- Click / Time spent on a particular retrieved document → implicit indication of relevance.

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- In absence of *true* feedback, assume top-ranked documents as relevant.

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- **Obvious danger**: if initial retrieval poor $\rightarrow$ feedback can aggravate the problem.
- Adhoc feedback tested to be useful on average.

Blind/Pseudo Relevance Feedback

Rocchio's algorithm

$$\mathbf{Q}_m = \alpha \mathbf{Q}_O + \beta \frac{1}{|D_R|} \sum_{\mathbf{d}_i \in D_R} \mathbf{d}_i - \gamma \frac{1}{|D_{NR}|} \sum_{\mathbf{d}_j \in D_{NR}} \mathbf{d}_j$$

- Perform a retrieval.

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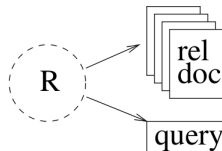
- Perform a retrieval.
- Consider top K ranked documents as **relevant** - D_R .
- Which documents to be selected as **non-relevant** - D_{NR} ?
- Only consider D_R and set γ to 0.

Relevance based Language Model (RLM)

Lavrenko and Croft - SIGIR 2001

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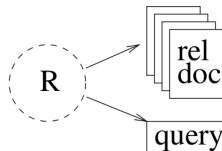
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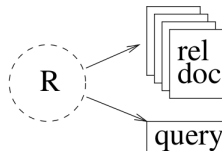
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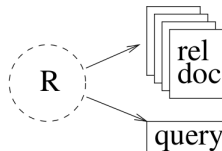
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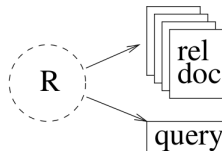
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- The task is to find (estimate) the density function for R .
- Estimates $P(w|R)$ on the basis of $P(w|Q)$.

RLM: Formally

PRP - a framework to define *'what is a good ranking?'*

- *If the response to a query is a ranking of the documents in the collection in order of **decreasing probability of relevance** ($P(R = 1|D, Q)$)*
 - *where the probabilities are estimated as accurately as possible*
- *... the overall effectiveness of the system will be the best that is obtainable.*

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PRP implies:

Rank documents by odds of their being observed in the relevant class.

$$\frac{P(D|R)}{P(D|N)} \sim \prod_{w \in D} \frac{P(w|R)}{P(w|N)}$$

The Estimation

- The probability of sampling a term w from the R , denoted by $P(w|R)$, approximated by

$$P(w|R) \approx \frac{P(w, Q)}{P(Q)}$$

- Joint probability of observing w along with query terms
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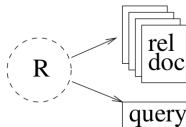
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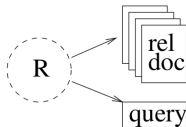
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- Given a query $Q = \{q_1, \dots, q_k\}$ and M , a **set of initially retrieved document**.

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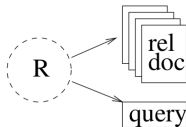


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- Query words $\{q_1, \dots, q_k\}$ and w sampled independently and identically from (pseudo-)relevant documents.

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- Query words $\{q_1, \dots, q_k\}$ and w sampled independently and identically from (pseudo-)relevant documents.
- Pseudo-relevant documents $\implies$ Top ranked documents.

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- $\prod_{q \in Q} P(q|D)$: Maximum likelihood estimate of Q in D (LM based retrieval score of D).
- $P(w|D)$: Maximum likelihood estimate of w in D .
- $P(D)$: Prior probability of selection of the document.

RM3

UMass at TREC 2004: Novelty and HARD - Jaleel et al. - TREC 2004

Drawback of RLM

Does not consider the original query terms for estimation of the density function

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RM3

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Given:

- Corpora : Set of documents.
- Query $Q : q_1, \dots, q_n$.
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- N Terms with high estimated value taken as expansion terms.
- Linearly interpolate derived density function with underlying query model:

$$P'(w|R) = \alpha P(w|R) + (1 - \alpha)P(w, Q)$$

Relevance Feedback: Summary

- Different types of feedbacks:
 - ▶ Explicit: Users do the work - **Expecsive**.
 - ▶ Implicit: System guesses based on user activity - **cheaper**.
 - ▶ Blind/Pseudo: Users not involved - **cheapest**.
- Query Expansion as a general mechanism.